DAM WATER CONSERVATION ASSISTANT

Using Machine Learning and MLOps

Detailed Project Report

A reservoir decision-support system for forecasting water availability, evaluating conservation scenarios, and improving the reproducibility of ML workflows.

Provisional reservoir candidate: K.R. Sagara (KRS) Dam, Karnataka

1. Abstract

Water conservation is an important challenge in reservoir management, especially during periods of low rainfall and reduced river inflow. Traditional monitoring focuses mainly on current water levels and storage, which may not fully explain future water availability or the benefits of conservation.

The proposed **Dam Water Conservation Assistant** uses historical reservoir data and machine learning to forecast future water availability and simulate conservation strategies. It analyzes storage, inflow, outflow, canal withdrawals and rainfall where available. Its main feature, **Water Survival Intelligence**, estimates how future availability may change under normal, low-inflow and conservation scenarios.

MLOps practices will manage datasets, track experiments, version models and monitor performance. The system supports decisions while respecting essential water requirements and official dam operating rules.

2. Introduction

Dams and reservoirs supply water for irrigation, drinking water, industry and other needs. Available water changes due to rainfall, river inflow, evaporation, releases and withdrawals. Machine learning can identify historical patterns and estimate future conditions. Combined with a water-budget engine, forecasts can help compare conservation options.

Unlike a simple water-level predictor, this project aims to explain how alternative water-use scenarios may affect future storage.

3. Problem Statement

Reservoir managers need to understand how much water is available now, how it may change, when storage could reach a reserve threshold, and how alternative withdrawal plans might affect future availability. These decisions are difficult when data contains gaps, inconsistent measurements or incomplete information about demand and operating constraints.

The project addresses this through forecasting, water-budget calculations, scenario simulation and data-quality monitoring.

4. Objectives

- Analyze historical storage, inflow, outflow and withdrawal patterns.
- Forecast future reservoir storage or inflow using machine learning.
- Calculate projected water availability with a water-budget model.
- Compare future availability under normal, low-inflow and conservation scenarios.
- Estimate potential effects of reducing or rescheduling eligible withdrawals.
- Flag missing, inconsistent or suspicious measurements.
- Present results through an interactive dashboard.
- Apply MLOps for data versioning, experiment tracking and model monitoring.

5. Proposed Solution

5.1 Data processing

Collect official reservoir records, select the chosen Karnataka reservoir, clean and validate observations, and standardize units.

5.2 Forecasting

Train models using historical storage, inflow, outflow, withdrawal and time-based features. Add rainfall when suitable catchment-level data is available.

5.3 Water-budget engine

Combine current storage with expected inflow, releases, withdrawals, evaporation and other transfers to estimate future storage.

5.4 Water Survival Intelligence

Compare a baseline plan with low-inflow and conservation scenarios, and estimate whether and when a defined reserve threshold may be reached.

5.5 Dashboard

Display observations, forecasts, water budgets, scenario comparisons and data-quality warnings.

6. Dataset Description

The project investigated the Karnataka daily reservoir dataset published through the National Water Data Portal. Important fields include:

Field	Description	Usage
Monitoring Date	Observation date	Time-series analysis
Reservoir Name	Reservoir identifier	Reservoir selection
Reservoir Level (ft)	Recorded level	Trend analysis and validation
Percentage Full	Storage percentage	Fullness analysis
Gross Capacity (TMC)	Gross-capacity field	Candidate storage target, pending validation
Live Capacity (TMC)	Live-capacity field	Storage analysis after definition verification
Inflow (Cusecs)	Water entering reservoir	Forecasting and budget
Outflow to River (Cusecs)	Water released to river	Water budget
Canal Withdrawal (Cusecs)	Canal diversion	Conservation scenarios
Evaporation (Cusecs)	Evaporative-loss field	Water budget when available
Cumulative flow fields (TMC)	Cumulative inflow, outflow, withdrawal	Consistency checks

The dataset does not include rainfall. A separate, suitable catchment-level rainfall source is needed for rainfall-aware forecasting.

6.1 Reservoir selection

Almatti was initially investigated, but missing evaporation data and gaps in daily observations limited complete water-budget analysis. K.R. Sagara (KRS) became the preferred candidate because its inflow, outflow and canal-withdrawal fields were highly complete.

Important: The KRS audit found suspicious storage and reservoir-level values. KRS remains a provisional candidate until storage definitions and anomalous records are verified against official sources. Reservoir selection should prioritize measurement reliability and water-budget consistency, not record count.

7. Data Preprocessing

- Filter the chosen reservoir while preserving the raw dataset.
- Parse dates, sort chronologically and investigate duplicate dates and missing calendar days.
- Convert storage, level, capacity and flow fields to numeric values; flag unparseable entries.
- Calculate missingness per column and handle values according to each model's requirements.
- Investigate negative flows, storage above design capacity, extreme levels and implausible jumps. Do not remove genuine floods or releases merely because they are unusual.
- Convert flow rates to volumes and verify gross/live storage definitions.
- Create lag features, rolling averages and seasonal variables.
- Split data chronologically and fit imputers using training data only.

Expected outputs are a validated reservoir dataset, a forecasting dataset and a water-budget dataset with quality flags.

8. Machine Learning Methodology

8.1 Inflow forecasting

Estimate future inflow using historical inflow, storage, past releases, temporal patterns and rainfall where available. Candidate models are a persistence baseline, Random Forest and optionally XGBoost. Use a more complex model only if it improves chronological test performance.

8.2 Storage forecasting

Estimate future storage from historical storage and flow-related features. Confirm the target's official definition before training; a field called gross capacity should not automatically be assumed to represent current stored water.

8.3 Evaluation

- MAE: average absolute prediction error.
- RMSE: error metric that penalizes large errors more strongly.
- Compare against a simple persistence baseline.
- Evaluate different forecast horizons and seasonal conditions.

Select the model using validation performance and report final metrics on an untouched test period.

9. Water-Budget Engine

The basic storage balance is:

S(t+1) = S(t) + I(t) – O(t) – W(t) – E(t) + T(t)

S is storage, I is inflow volume, O is river-release volume, W is canal withdrawal, E is evaporation loss and T is net other transfers. All terms must use compatible volume units.

For a flow rate in Cusecs representing the mean flow over a complete day: **Daily volume (TMC) = Mean Cusecs × 0.0000864**. Shorter or instantaneous measurements require appropriate integration. Compare calculated and observed storage changes; flag significant residuals instead of automatically interpreting them as leakage.

10. Water Survival Intelligence — Main Novelty

This feature extends beyond predicting water levels by estimating how future reservoir availability changes under alternative hydrological and water-use scenarios.

10.1 Scenarios

- Baseline: specified planned releases and withdrawals with forecast inflow.
- Low inflow: simulate a lower-inflow condition.
- Conservation: reduce or reschedule eligible withdrawals hypothetically.
- Combined stress: combine lower inflow with an alternative demand assumption.

Scenarios are estimates, not guarantees of future hydrological conditions.

10.2 Reserve-threshold estimation

The user defines a threshold based on approved operating requirements. The simulator estimates the threshold date if reached within the forecast horizon and reports projected storage, scenario differences, assumptions and uncertainty. It must not claim a reliable number of days remaining unless material inflows, releases, withdrawals, evaporation and transfers are accounted for.

10.3 Conservation benefit

Compare the baseline with a conservation scenario to estimate storage differences and any change in the threshold date. The system must not override essential supplies, environmental flows, flood-control requirements or official operating

constraints.

11. Data Quality Monitoring

Monitor missing or delayed observations, implausible measurements, capacity inconsistencies, large water-balance residuals, changes in feature distributions and forecast errors. Flag suspicious observations for review; do not automatically treat them as evidence of a dam leak.

12. MLOps Architecture

Data ingestion → Validation → Preprocessing → Feature engineering → Training → Evaluation → Model registration → Forecasting → Water-budget simulation → Dashboard → Monitoring

12.1 DVC — Data versioning

Track dataset versions, preprocessing outputs and pipeline dependencies alongside Git to make experiments reproducible.

12.2 MLflow — Experiment tracking

Track parameters, metrics, experiment runs and model artifacts; compare models and feature configurations.

12.3 Monitoring

Track forecast errors, data quality and distribution changes. Raise warnings when performance degrades or inputs become unreliable.

12.4 Reproducibility

Keep consistent configurations, data versions, model versions and preprocessing logic between training and inference.

13. Proposed Technology Stack

Component	Technology	Purpose
Programming	Python	Data processing and ML
Data analysis	Pandas, NumPy	Cleaning and transformation
Visualization	Matplotlib or Plotly	Analysis and plots
Machine learning	Scikit-learn; optionally XGBoost	Forecasting
Experiment tracking	MLflow	Experiment history and metrics
Data versioning	DVC with Git	Dataset and pipeline versioning
Dashboard	Streamlit	Interactive interface
Storage	CSV initially; SQLite if needed	Data and application records

14. Dashboard Features

- Reservoir overview: storage, fullness, inflow, outflow and withdrawals.
- Forecasting: future storage/inflow and model performance.
- Water budget: projected storage and accounted water flows.
- Scenario simulator: baseline, low-inflow and conservation cases.
- Water Survival Intelligence: reserve-threshold estimates and scenario differences.
- Data quality: missing values, suspicious readings and balance warnings.
- Model monitoring: model version, metrics, prediction errors and monitoring status.

The interface must distinguish observed measurements, ML predictions and hypothetical scenario outputs.

15. Testing and Evaluation

Data tests

- Validate dates and column types.
- Check missing-value handling.
- Test capacity relationships and unit conversions.
- Detect storage jumps and cumulative-counter resets.

Model tests

- Compare ML against persistence baselines.
- Use chronological validation and test sets.
- Evaluate across seasons and storage conditions.
- Prevent future information from leaking into features.

Water-budget tests

- Compare calculated and observed storage changes.
- Investigate large residuals.
- Verify compatible units.
- Test the impact of missing measurements.

Application tests

- Verify dashboard inputs and outputs.
- Compare scenario calculations with manual examples.
- Warn on invalid or incomplete inputs.

16. Expected Outcomes

- Cleaned and versioned reservoir dataset.
- Reproducible preprocessing pipeline.
- Validated baseline and selected forecasting model.
- Water-budget calculation engine.
- Conservation scenario simulator.
- Water Survival Intelligence for reserve-threshold estimates.
- Interactive dashboard.
- MLOps workflow for tracking experiments, models and data quality.

17. Limitations

- Historical observations alone cannot fully determine future water availability.
- Rainfall is absent from the current reservoir dataset and must be sourced separately.
- Demand, authorized releases and operating constraints may require additional sources.
- Missing evaporation and unmeasured transfers can affect water-budget accuracy.
- Forecast uncertainty increases with horizon.
- Suspicious storage readings must be verified before model training.
- Scenario results depend on assumptions and do not guarantee future conditions.
- The system supports decisions and does not replace official operating procedures or engineering judgement.

18. Future Enhancements

- Integrate catchment-level rainfall forecasts.
- Use upstream river observations for inflow forecasting.
- Add uncertainty intervals for multi-day predictions.
- Automatically flag suspicious water-budget residuals.
- Compare conservation scenarios under multiple demand assumptions.
- Integrate verified reservoir operating rules.
- Automate data-quality reporting and retraining workflows.
- Extend to other Karnataka reservoirs after validating their data definitions and quality.

19. Conclusion

The Dam Water Conservation Assistant combines machine learning, water-budget calculations and MLOps practices to help analyze reservoir conditions and evaluate conservation strategies. Water Survival Intelligence compares future availability under alternative inflow and withdrawal scenarios.

KRS is the provisional reservoir candidate, subject to storage validation. DVC and MLflow will support reproducibility and model tracking, while a dashboard will communicate predictions and assumptions.

The final system will provide estimates, comparisons and data-quality warnings to support conservation decisions. Its usefulness depends on validated measurements, forecasting performance and reliable demand and operating information.